# Teaching Statement

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The field of computer science is a fast-paced discipline characterized by rapid technical developments and continual evolution. Thus, it is important for students to understand the foundations of this field and develop critical-thinking and problem-solving skills to be able to keep up with advances. I strongly believe that these can be achieved through thoughtful teaching methods, interactive learning, and effective mentoring approaches. My teaching philosophy is rooted in preparing students not only to understand current technologies, but also to adapt to the innovations that they will encounter in the future.

## Teaching Philosophy

I am dedicated to designing and teaching courses that **expose students to foundational principles and emerging technologies**. I believe that this is applicable for both undergraduate courses (by connecting the topic under study to real life applications and industry practices) and advanced graduate courses (by encouraging students to investigate current developments and open problems). Despite the abundance of educational resources available today, it can be difficult for students to identify comprehensive and effective resources for learning the foundations. This challenge can be overcome through carefully designed courses and effective pedagogical strategies that allow students to grasp core concepts and develop the skills needed for independent learning and growth. I invest time and effort to ensure that my courses meet such criteria.

Moreover, **active student engagement** is a central component of my teaching approach. In my experience, an interactive class style helps students to learn the course subject, and helps the instructor to determine which topics need more emphasis and what the students' concerns are. To achieve this goal, I encourage students to ask questions inside and outside the classroom, lead in-class discussions, and utilize online and in-person platforms to facilitate interaction.

Furthermore, I believe that providing **continuous assessment and support** for students is invaluable. To satisfy this criterion, I invest time and effort in developing well-thought assignments, projects, and exams. These assessments help in evaluating the level of students' understanding of the course concepts and their progress, motivating them to apply newly acquired knowledge to solving challenging problems, and learning how to articulate and communicate their thoughts.

Lastly, I believe it is very important to **encourage students to collaborate with each other as well as work independently**. Becoming a team player, and being able to find the right collaborators, makes the learning process more productive and enjoyable. At the same time, the ability to solve problems individually enables assessing the level of understanding and spotting any lack or deficit. Thus, I develop assignments and projects that align with this goal.

## Teaching Experiences and Future Plans

My teaching experience spans undergraduate and graduate education, curriculum development, and research mentoring across multiple institutions. After my undergraduate degree, I worked as a lab supervisor in the Electrical and Computer Engineering Department at the Hashemite University, where I taught Digital Logic Design and Microprocessors labs. Then, after finishing my masters, I joined the Computer Engineering Department at the Hashemite University as a faculty member. I taught basic and advanced undergraduate courses, many of which required developing the course material. These include Digital Logic Design, C++ Programming, Data Structures, Modeling and Simulation, and the Computer Maintenance lab.

During my PhD studies, I served as a teaching assistant for the Introduction to Cryptography course. I was also a PhD coordinator of the Emerging Scholar Program (ESP), a seminar series for undergraduates to learn about the sub-fields of Computer Science (CS). My role involved planning the seminars, reviewing their materials, and holding guest speaker sessions. Furthermore, I was invited as a guest lecturer to talk about my research in several cryptography and security courses at Columbia, NYU, and Fordham University.

After joining UConn, I developed and regularly teach Blockchain Technology, a research-oriented course that teaches students the core technical concepts of this emerging technology, and explores recent academic and industrial advances. I also teach the Introduction to Cryptography and Cybersecurity course (cross-listed as graduate-undergraduate course). This course covers the fundamentals of cryptography and network security, and aims to enable students to develop the cybersecurity mindset needed for secure systems design. These experiences have shaped my approach to curriculum development, course design, and student engagement.

**Future course offering plans.** Looking ahead, I plan to continue expanding the department's offerings through courses that connect emerging technologies with societal and security challenges. I plan to develop two courses: (1) Blockchains and Social Good, which is inspired by my work on decentralized systems and how blockchains may help in creating equitable/transparent services and resource markets. The course will be open for both CS and non-CS students and will explore a variety of topics covering the societal, environmental, legal, and financial impacts of blockchains and cryptocurrencies. (2) Privacy for Emerging Models, which is inspired by my work on privacy-preserving cryptographic solutions such as multiparty computation, homomorphic encryption, etc. The course will first introduce these technologies, and then explore their use in, e.g., private/anonymous cryptocurrencies, privacy-preserving machine learning, etc.

## Mentoring

I am committed to continuing and expanding my efforts in advising and mentoring students. Supporting students as they develop into independent researchers and professionals is one of the most rewarding aspects of my academic career.

Going back to my first academic job, I worked closely with undergraduate students by supervising their senior design projects and motivating them to participate in international and national competitions. Now at UConn, I regularly collaborate with undergraduate, master, and PhD students on research projects. I also take every opportunity to offer advice regarding their career paths, and provide insights based on my entrepreneurial, academic, and research expertise.

The mentoring/advising nature is different for undergraduates versus graduates. For undergraduates, the goal is often to introduce students to research and help them to determine whether research is a path they wish to pursue further. Projects for undergraduates are well-defined with a clear description of the student's role and task. I have mentored and collaborated with several undergraduate students. In addition to contributing to publications, two of these students chose to pursue graduate school (one joined my group as MSc student and another joined another institution as a PhD student).

For graduate students, my mentoring approach is initially hands-on and gradually transitions into giving students more freedom so they can learn how to become independent researchers. I believe that motivating students to find research problems and questions that interest them is essential. I also encourage them to attend technical conferences and workshops, interact with other researchers, give talks, etc. The goal is that by the end of their PhD studies, students can independently formulate research questions, develop and evaluate solutions, communicate their findings effectively, and collaborate with other researchers in a productive way. My support will continue even after they graduate, and I will do my best to offer advice, collaborate on research projects, and push them to do independent work and find their own collaborators. I currently advise/have advised five PhD students and two MSc student, and I have served on the committees of several PhD students in the department.